

Analysis of the Performance of a Bike-Share Rental Company in 2017/18

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Executive Summary

This report has been conducted to analyse the performance of a Bike-Share Rental Company in Seoul, spanning the period of December 2017-November 2018. The company provides a service for customers to rent and share a bike they can use to cycle around the city.

This company has provided data on the daily number of rentals, in addition to a range of weather data per day. The provided data has shown that there is a very clear cyclical trend of popularity of bike rentals, with significantly fewer bike rentals during certain seasons compared to others. The weather data has also shown that the weather does have a noticeable impact on bike rental rates in Seoul.

Some limitations arose during the analysis. Many of the weather variables may have intervened with the analysis, as there were many codependent/interdependent variables. Perhaps they could consider measuring fewer, but more relevant variables- for example visibility and humidity (which didn't seem to have much effect) could be replaced with simply measuring temperature and rain.

Another recommendation would be to include data more relevant to the operation and management of the business. The weather is not something the company can adjust or change, so there is a limitation to the usefulness of the statistics. One recommendation would be to include a variety of information other than weather- such as internal business factors such as profit and loss, as this would provide the company with metrics they can actively change and work on, so that they can satisfy their objectives. Additionally, external factors such as tourism rates, can also be an important one to consider.

1. Introduction

1.1 Background on the Study and the Company

All companies should consider if they have any seasonal trend in popularity, as this can help them adjust their objectives during the year. Seasonality may affect different companies differently, and some may not have any. This can be dependent on the nature of a business, or naturally cyclical events such as weather.

Seoul is a city with a temperate climate with four distinct seasons- spring, summer, autumn, winter. Seoul's climate means that the differences between summer and winter can be very extreme. As a result, this can have a heavy impact on services such as bike rental companies, where customers and tourists rely on weather when they decide on their daily activities. Therefore, it is especially for the company to consider this.

1.2 Aims of the Analysis and its' Potential Influence

The aim of this report is to show how significant the impact of different variables, such as wind, rain, and solar radiation is, on the amount of interested cyclists. The data provided by the company has been used to visualise seasonal trends, as well as identify the impact of different weather variables on bike rental rates. Using this analysis, the company can determine timings of certain business decisions such as staffing or inventory, to more efficiently allocate resources. Additionally, this report can show how efficiently they are already operating.

1.3.1 Methodology

The analysis was done using analytical and graphical linear regression. Regression data, specifically R numbers and P numbers were identified, and were used to consider whether certain variables were significant or not. Multiple linear regression involved grouping all (or some) of the variables, to see the significance of the data sets as a whole. Graphical analysis showed the regression data visually, so that it can be understood more quickly and clearly.

Seasonal trends were identified using trendlines with varying moving average periods, to smooth out irregularities and anomalies. These anomalies were caused by irregular weather, as well as the fact that there were 13 non-operational days out of the 366 days. This is shown in *Figure 1*, where the red trendline has a 34-period moving average. This was sufficient to show a clear trend, without compromising too much of the data.

1.3.1 Methodology Evaluation

Since there was data for every day over the course of a year, there was a sufficient sample size to conduct accurate multiple linear regression. However, not all regression models were appropriate, and some violated some assumptions for regression modelling. Some variables had multicollinearity, which meant the effect of individual variables on the companies' performance could not be clearly separated. This may have artificially inflated the statistical significance of certain variables. As a result, not all variables were considered equally.

Further evaluation on the appropriateness of the data and regression models will be discussed later in the regression analysis section of this report (sections 2.x).

1.4 Findings- Seasonality

The findings show there is a clear seasonal trend, with less bike rentals during winter and spring months, and a steep increase from summer through autumn. This is likely to be due to less bicycle-friendly conditions due to rain, wind and/or snow in certain times of the year, and dry, sunny weather bringing in more outdoor enthusiasts. (More in-depth analysis of the impact of certain weather variables will be included later in this report, in the regression analysis sections 2.x).

The peak happens to be during summer holiday season around July/August 2018 (as shown in *Figure 1*), which could be caused for example by a surge of tourism into Seoul; however, data on tourism was not provided.

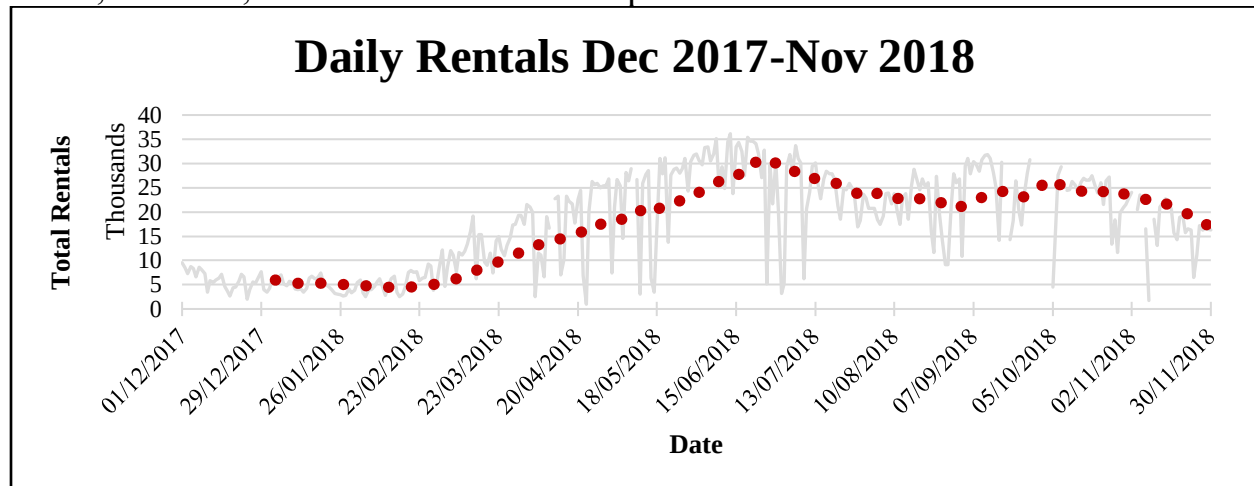


Figure 1

The seasonal trend also shows that on average at the start of the yearly cycle there was more bike rentals than at the end. The first week average was 8,220, and the last week average was 14,340, with a percentage increase of 74% (shown in dashboard). This could mean one of two things- either that there is overall growth in the company, and they are operating and managing efficiently, or the seasonal cycle lasts longer than a year, and this data does not represent a full completed cycle. However, this is

unlikely since this is very uncommon- a longer period of data would be needed to see if there is a further repeating trend.

2. Regression Analysis- Using Only Quantitative Data (Weather Variables)

2.1 Findings

Figure 2 shows the multiple linear regression analysis of the impact of all the weather variables on daily bike rental rates.

<i>Regression Statistics</i>	
Multiple R	0.812254663
R Square	0.659757638
Adjusted R Square	0.652133159
Standard Error	6064.72507
Observations	366

<i>ANOVA</i>					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	8	25461647075	3.183E+09	86.53150779	7.54042E-79
Residual	357	13130777791	36780890		
Total	365	38592424866			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>
Intercept	17501.04148	13015.40796	1.3446403	0.179595124
Avg. Temp	-32.072635	502.7332824	-0.0637965	0.949167953
Avg. Humidity	-106.7897464	147.3531946	-0.7247196	0.469098785
Avg. Wind	-1918.594827	587.4467174	-3.2659895	0.001196485
Avg. Visibility	0.57378706	0.863680229	0.6643513	0.506894376
Avg. Dew Point Temp	479.4002957	532.5650159	0.9001723	0.368635354
Avg. Solar Radiation	12522.91695	1716.379661	7.2961229	1.93284E-12
Total Rain	-151.6566227	35.50429676	-4.2715005	2.49288E-05
Total Snow	-28.27474022	39.24764941	-0.7204187	0.471738678

Figure 2

2.1.1 R Squared Value- Evaluation

Firstly, the R Square value given is 0.66. this means that the model including all the weather data accounts for about 66% of the bike rental data, or that the correlation value is 66%. This shows that there is a noticeable correlation between the weather data and bike rentals, although it is not very strong. About 34% of the bike rental data must be explained by other factors. This must mean that either there are factors that have not been measured, or that some of the existing factors may be insignificant enough to be ruining this model.

2.1.2 F value- Evaluation

The significance F value in this case is practically 0.00. This highlights that there is in fact a statistically significant correlation in this model, and the results may not have occurred by chance.

2.1.3 Coefficient- Evaluation

The coefficient is one way of seeing how significantly a variable affects the bike rental rates. Some variables have negative or positive coefficients. For example, if average dew point temperature were to go up by one unit one day, it is predicted that bike rental rates would be about 479.4 higher that day since it has a positive coefficient. On the other hand, if rain was to be increased by one unit, this would decrease the bike rental rates by 28.27. Average visibility has a coefficient close to 0, which would mean a relatively big change in visibility wouldn't have a very significant impact on bike rentals. As a result, this variable may be ignored, to create a more accurate regression model.

However, this poses an issue- some variables have extremely sensitive/insensitive units. In the data set provided, average visibility values range from 214.29 to 2000, which is a very broad range of values, whereas solar radiation ranges from 0 to 1.22. in practice, the difference between and 1.22MJ/m² of solar radiation is night and day, whereas the difference between 200 and 201.22 visibility is unnoticeable. Because of this, either the units must be adjusted, or these variables must be ignored to improve the accuracy of the coefficient of this model.

2.1.4 P value- Evaluation

P value is another statistical test to see how significant a variable is. In this model the only statistically significant variables, with a P value under 0.01 (which is generally considered good), are wind, solar radiation, and rain. This questions the appropriateness of this model, as this means that most of the other variables should be ignored from this model, as it means that it must violate some assumptions for regression modelling.

One of the assumptions this violates is multicollinearity. Logically, a variable like temperature, which has a very high P value, will have a strong correlation with others such as snow, or dew point temperature. This means that the effect of the individual variables cannot be clearly separated, so the regression equation becomes unstable. Another assumption broken is homoscedasticity. Simple linear regression graphing (*figure 3*, *figure 4*) shows that some variables such as dew point don't follow homoscedasticity and are in fact heteroscedastic (*figure 3*).

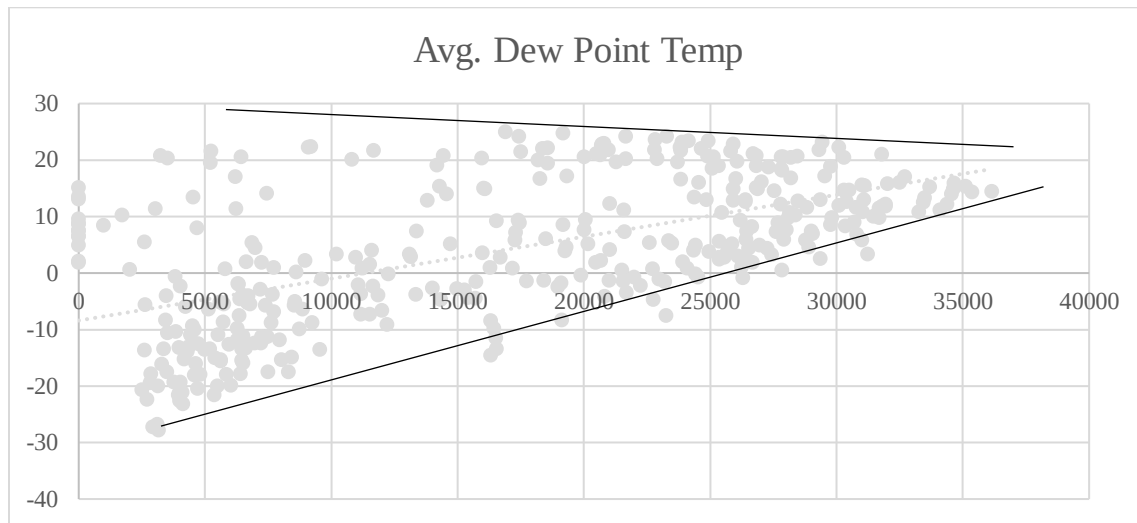


Figure 3

This is opposed to solar radiation, which follows a homoscedastic pattern (*figure 4*).

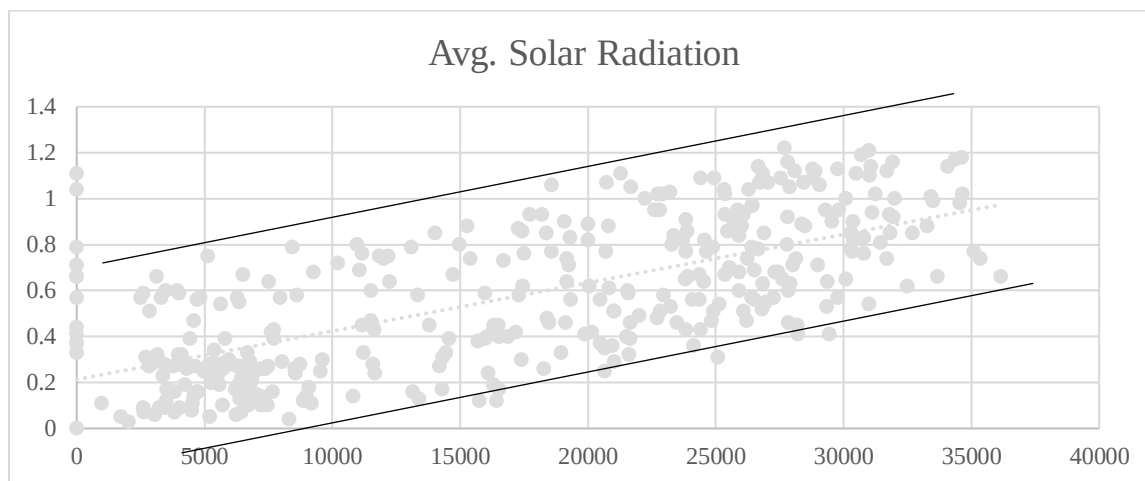


Figure 4

3. Expanded Regression Analysis- Accounting for Qualitative Data (Seasons Included)

3.1 Findings and Evaluation

Regression Statistics		
Multiple R	0.82354087	
R Square	0.67821956	
Adjusted R Square	0.67008467	
Standard Error	5906.16876	
Observations	366	
ANOVA		
	Significance F	
Regression	3.8846E-82	
Residual		
Total		
	Coefficients	P-value
Intercept	26615.5675	0.038822033
Season	5297.67022	8.45086E-06
Avg. Temp	-494.558005	0.323442559
Avg. Humidity	-238.298361	0.104518994
Avg. Wind	-1765.8515	0.002221467
Avg. Visibility	0.31365946	0.710080252
Avg. Dew Point Temp	864.961312	0.100722682
Avg. Solar Radiation	10783.1218	9.49932E-10
Total Rain	-153.44312	1.21488E-05
Total Snow	-3.22604527	0.933476937

figure 5

(Some irrelevant statistical values have been removed from *figure 5* and *figure 6*).

These findings in *figure 5* show that with the addition of seasons as a dummy variable, where spring, summer and autumn take the value 1 and winter takes 0, we can see some minor differences in this model compared to the first:

- The R squared value has increased from 0.66 to 0.68 which is an improvement, and the significance F value has decreased even further, although this is a negligible difference.
- The P value for the season variable is significantly below 0.01, which means that the seasons have a great influence on bike rentals. This makes sense, as only 7.89% of all bike rentals were in winter.
- The addition of another variable has changed the P-values of all the other variables, however there are still only three others with a value below 0.01, which are the same ones as the first model

Adding too many variables and data to a multiple regression analysis may complicate the regression formula and deem it insignificant. In addition, seasons are also highly correlated to many of the other variables, which violates the requirements for proper regression modelling. Therefore, it is best to ignore this.

4. Improved Regression Model

4.1 Findings and Evaluation

<i>Regression Statistics</i>		
Multiple R	0.811694063	
R Square	0.658847251	
Adjusted R Square	0.654109019	
Standard Error	6047.476935	
Observations	366	

<i>ANOVA</i>		
	<i>Significance F</i>	
Regression	8.40653E-82	

	<i>Coefficients</i>	<i>P-value</i>
Intercept	18889.92948	2.41E-10
Avg. Humidity	-124.33992	0.0027
Avg. Wind	-1823.52029	0.001581
Avg. Dew Point Temp	467.6080455	1.11E-18
Avg. Solar Radiation	12283.55809	7.13E-13
Total Rain	-147.527125	1.09E-05

Figure 6

Figure 6 shows an improved multiple regression model, with some of the variables with too high P-values removed. This is because those variables have shown to be statistically insignificant in both previous models, meaning they didn't have enough of an impact on bike rentals. This leaves all the remaining variables with highly significant P-values way below 0.01.

Less variables also decreases the risk of multicollinearity, and the remaining variables have a lot less co-relation between them. As a result, this makes this model more in line with the assumptions of regression modelling. the significance F value is also still just as low as previous models.

However, the R square value is still very similar to the previous models, despite adding and removing variables. This must mean that weather generally only explains roughly 65% of the company's bike rental data, and the remaining 35% is due to

extraneous variables. A recommendation for future reports would be to include other data, such as internal business figures, or external data such as tourism numbers.

5. Appendix

5.1 Dashboard summary and insights

The main part of this dashboard is the time-series chart of daily rentals over time, showing the total rentals per day (gaps in the graph show non-operating days for clarity). This chart can be separated by season using the Slicer toggle to the right of it. The graph clearly shows the seasonal trend and can be used to identify the best months for business.

The best days for business are shown in the funnel chart on the right, showing the top 5 highest performing days- which are all, interestingly, in June. This could be used, for example, to reward staff working on those days. The funnel chart below it shows the days with least bike rentals, which can be due to anomalies or a quiet season. Perhaps these anomalous days could be investigated, to see what caused them to learn and improve.

Similarly, the non-operating days table could be investigated, to see why some days the company wasn't operating- whether it was a holiday or not. The sunburst chart visually shows how many days of the year the company was operating or not.

Seasons are an important factor to consider for this company, and it is also important to know the exact number of sales for any company, to calculate revenues, costs, and profits. There are charts showing the daily average per season, as well as the total number of rentals per season, which can be used to calculate which seasons are most profitable.

The overall yearly total is shown in the table at the top, so that the company can also calculate their total profits in the given year. It also shows the growth of the company in the year, which is calculated by the percentage change from the start of the year to the end of the year. This shows the overall trajectory the company is going in and reveals momentum going into the next year.